

ASTRUM DRIVE

ASTRA

Compact, Verifiable AI for Scientific Discovery

A multi-model research platform that turns scientific objectives into falsifiable conjectures, reviewed executable tests, reproducible evidence, and the next decision.

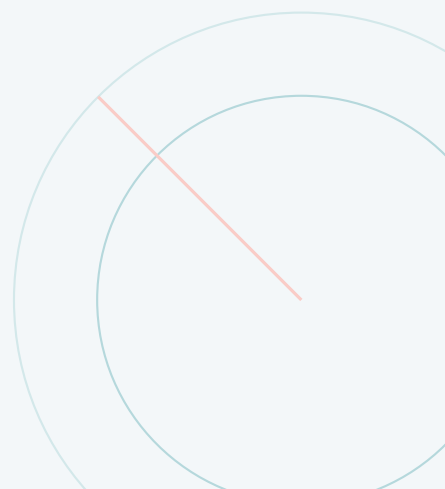
Investor, VC, Client and Strategic Partner Evidence Brief

Product thesis, current benchmark evidence, competitive reference architecture, algorithmic workflow, deployment economics, commercialization roadmap and explicit limits.

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Founder and Lead Architect | Astrum Drive

July 2026 | Discussion document



Executive thesis

ASTRA makes AI-generated scientific reasoning inspectable.

Most AI systems optimize the answer. ASTRA optimizes the evidence chain: independent models propose and challenge a claim, a specialist writes a falsifiable validator, another model audits that validator, computation engines execute it, and a human controls what becomes accepted knowledge.

The problem

Scientific and engineering teams can generate hypotheses and code faster with AI, but the bottleneck moves downstream: deciding whether the produced result actually tests the intended claim. Plausible prose, a successful program, or a printed PASS is not sufficient evidence.

This creates four practical risks:

- ▶ hidden changes in assumptions, domains or boundary conditions;
- ▶ validators that confirm their own construction;
- ▶ numerical checks presented as universal proofs;
- ▶ fragmented evidence across chats, notebooks and tools.

The ASTRA response

ASTRA coordinates complementary AI models and deterministic scientific engines around one shared objective. Generation, code authorship, code review, execution, result analysis and research navigation are deliberately separated.

The product is designed to produce:

- ▶ a precise and falsifiable statement;
- ▶ executable validation code with explicit failure paths;
- ▶ raw output from symbolic, numerical or logical engines;
- ▶ a conservative interpretation tied to the original objective;
- ▶ a persistent record that experts can reproduce and challenge.

Investment proposition. ASTRA is an orchestration and assurance layer for AI-assisted R&D. It is model-flexible, engine-extensible and deployable close to private data and scientific software. The near-term opportunity is to convert an operational research system into a pilot-ready product for teams that need faster experimentation without surrendering traceability.

Category reference. Google DeepMind's Deep Think, Aletheia, Co-Scientist and AlphaEvolve demonstrate that long-horizon search, debate, verification and automated evaluation can extend frontier models in science. ASTRA adopts that general direction but makes a different product choice: heterogeneous models, reviewed executable validators, local or private compute, and evidence bundles that can be inspected by a customer.

What exists today

68/68

automated production tests passing

8/8

local-ASTRUM client evidence bundles passing

100%

cross-oracle agreement on four paired cases

The current system includes a browser application, an MCP interface for agent control, local and remote execution, detached long-running jobs, benchmark suites, report persistence, formal proof and CAS routes, package extensions, bounded repair and human approval gates. Validator Repair vNext.1 compiles and audits code before model review, applies safe source-local fixes in milliseconds and restricts Claude to one exact, auditable patch rather than a wholesale regeneration.

Evidence position, not a leaderboard claim. ASTRA currently demonstrates operational reliability, cross-environment reproducibility and useful multi-model controls. In a four-case paired public-benchmark canary, the heterogeneous proposal team produced greater measured perspective separation than the homogeneous control (0.7459 versus 0.5958), but both passed 2/4 cases. It does not yet demonstrate higher scientific success than Deep Think, Aletheia or other frontier systems. The frozen long-horizon canary is the release gate for that stronger claim; the first vNext.0 result was a valid null result.

Part I – Product and commercial view

One product, three audiences

Audience	Immediate value	Typical entry point
R&D teams	Move from a research question to a reproducible computational test while preserving assumptions and failed paths.	Theoretical physics, symbolic models, differential equations, optimization, simulation validation.
Engineering clients	Audit AI-generated scientific code and compare results across independent engines or internal packages.	Model verification, solver regression, design-space exploration, mathematical QA.
Investors / partners	Back an assurance layer positioned between frontier models and high-value scientific workflows.	Design partnerships, vertical modules, enterprise deployment, compute and model integrations.

The customer job

When a technical team has a consequential scientific claim, **it needs** to translate that claim into a test that can fail, execute it with the right tools, and preserve enough evidence for another expert to reproduce the decision. **ASTRA packages that workflow as a governed research cycle.**

Candidate use cases

- ▶ symbolic identities, invariants and limiting cases;
- ▶ general relativity metrics and curvature calculations;
- ▶ quantum operators, states and consistency checks;
- ▶ differential equations and boundary-value problems;
- ▶ fluid mechanics and dimensional validation;
- ▶ counterexample search with Z3;
- ▶ formal logic and Lean 4 proofs;
- ▶ optimization using internal scientific packages;
- ▶ cross-validation through Mathematica;
- ▶ reproducibility audits for model-generated code.

Value creation

Faster path to evidence

Automates the transitions between hypothesis, formalization, executable test, correction and interpretation.

Reusable organizational knowledge

Stores validated results, refutations, scripts and reports instead of losing them in isolated conversations.

Lower verification friction

Places a code review and deterministic execution between model output and scientific acceptance.

Flexible scientific stack

Routes work through local, remote, open-source, formal and in-house computation engines.

Commercial models to validate

The architecture supports several commercial forms; these remain go-to-market hypotheses rather than declared pricing:

- ▶ individual and team subscriptions for research orchestration;
- ▶ enterprise licensing for private deployment and governed evidence retention;
- ▶ paid vertical modules for specific scientific domains;
- ▶ integration, validation and joint research programs with internal software, compute, model and scientific-software partners.

Why ASTRA can be differentiated

	Conventional AI assistant	ASTRA
Primary output	A response or code artifact	A traceable evidence chain
Model structure	Often one model and one context	Multiple models with separated responsibilities
Validation	Optional, user-initiated	Mandatory execution in the core cycle
Code quality control	Same model may author and judge	Independent reviewer can reject and request revision
Scientific engines	Ad hoc tool use	Routed oracle with explicit engine metadata
Failure semantics	Errors may be folded into prose	REFUTED, CODE_ERROR, API_ERROR, weak evidence
Knowledge governance	Chat history	Reports, branches, axiomatic base and human gate
Deployment posture	Primarily hosted	Local-first with optional remote execution

Defensibility thesis

Workflow IP and calibration. The valuable asset is not a single prompt. It is the evolving set of role contracts, review rules, verdict guards, correction behavior, benchmark cases and domain-specific validation patterns.

Engine and package graph. ASTRA can accumulate verified adapters and test protocols across symbolic algebra, SMT, numerical science, formal proof, Mathematica and proprietary packages.

Evidence and failure data. With appropriate customer consent and governance, recurring validator defects, refutation patterns and correction outcomes can improve system calibration.

Model independence. The orchestration layer can adopt stronger models without rebuilding the scientific workflow. Models are replaceable capabilities; the evidence architecture persists.

Reference landscape: frontier research agents

The category is moving from one-shot answers toward compound systems that search, debate, verify and evolve candidate work. Google DeepMind reports professional mathematics and science results from Gemini Deep Think and its Aletheia research agent.¹ Aletheia iteratively generates, verifies and revises natural-language solutions.² Google's Co-Scientist uses a multi-agent generate-debate-evolve process for hypothesis generation,³ while AlphaEvolve combines a fast and a deep model with automated evaluators and evolutionary search over programs.⁴

¹Google DeepMind, "Accelerating Mathematical and Scientific Discovery with Gemini Deep Think," 2026.

²Feng et al., "Towards Autonomous Mathematics Research," arXiv:2602.10177, 2026.

³Google DeepMind, "Co-Scientist: A multi-agent AI partner to accelerate research," 2026.

⁴Google DeepMind, "AlphaEvolve," 2025.

System	Publicly strength	described	Relationship to ASTRA
Gemini Deep Think / Aletheia	Frontier reasoning plus a long-horizon generator-verifier-reviser loop for professional mathematics and scientific problems.		Closest epistemic reference. ASTRA does not claim comparable published research performance; it externalizes verification into reviewed code and customer-controlled engines.
Google Scientist	Co-Asynchronous specialized agents generate, debate, rank and evolve research hypotheses with scientist guidance.		Validates multi-agent scientific collaboration as a category. ASTRA uses a smaller fixed role map and emphasizes executable evidence after hypothesis generation.
AlphaEvolve	An ensemble balances breadth and depth while automated evaluators drive evolutionary program search.		Shares evaluator-grounded iteration. ASTRA targets heterogeneous scientific workflows rather than domains with one cheap scalar evaluator.
ASTRA	Codex, Antigravity and Claude separate conjecture, navigation, implementation and review; deterministic oracles decide executable checks.		Local-first assurance layer for physics, mathematics and engineering, with package adapters, human approval and auditable failure semantics.

The shared principle and the distinct product bet

Shared principle. Verification is a process rather than a final prompt. Candidate work must be challenged, revised or discarded; proof and refutation should be pursued together; tools can extend reasoning; and an expert remains responsible for consequential acceptance.

ASTRA's product bet. Scientific customers also need model plurality, validator-code review, engine-grounded evidence, private deployment, internal-package integration and an organizational approval trail. ASTRA packages those controls without training a new foundation model.

No false equivalence. Deep Think, Aletheia, Co-Scientist and AlphaEvolve are frontier research programs with reported results that ASTRA has not reproduced. Their public disclosures do not provide an apples-to-apples per-discovery cost. ASTRA's current claim is narrower: useful controls and reproducible evidence can be delivered with a compact three-model orchestrator and a single scientific workstation.

Benchmark evidence as of July 26, 2026

What has been measured

Evidence gate	Result	Interpretation
Automated production suite	68/68 passed	Core routing, deliberation, Lean safeguards, external evaluators, trajectory metrics, validator repair and client evidence remain regression-tested.
Local client validation	5/5 passed	Z3, SciPy, Pint, SymPy and GR_python produced complete evidence bundles in the production Python environment.
Local-ASTRUM replication	8/8 passed; 100% agreement	Four application cases produced matching claim verdicts on Windows and the remote Linux workstation. This is reproducibility evidence, not broad scientific accuracy.
Parallel execution smoke	8/8 grade-A runs	Four deterministic cases ran on both oracles with four workers: verdict accuracy and cross-oracle agreement were both 100%; P50/P95 latency was 5.422/8.598 seconds.
vNext.1 local repair gate	4/4 repaired	Every exact source pattern that the deterministic repairer claims to support was detected and repaired; median/P95 repair time was 0.702/1.913 ms.
Sound-code calibration	0/3 false blocks	Three sound validators, including an EinsteinPy API regression, remained approved by the deterministic gate. The sample is intentionally small and must grow.
Paired diversity canary	2/4 passed in both arms	Heterogeneous proposers increased mean perspective separation from 0.5958 to 0.7459, but produced no pass-rate uplift in four public cases.
Long-horizon vNext.0 canary	0/4 credible cycles	The first repair strategy did not improve conversion and averaged 1,045 seconds per cycle. ASTRA records this as a null result, not a success.
vNext.1 one-cell diagnostic	reviewer timeout; repaired artifact 12/12	The maximum-model cycle stopped at the 240-second reviewer ceiling after 1,451 seconds. A diagnostic ASTRUM run exposed one noncanonical symbolic-zero failure; the new deterministic repair made all twelve checks pass. This remains unreviewed diagnostic evidence, not an admitted scientific result.

Why the null results are useful. vNext.0 showed that asking Claude to regenerate large validators more often increased latency without clearing the reviewer. vNext.1 therefore applies safe deterministic edits first, performs compile/import smoke checks, allows at most one bounded exact-edit model patch, and refuses scripts too large for that patch context. The one-cell vNext.1 diagnostic then added a third local repair: canonicalization of raw symbolic tensor-zero checks. This is a measured improvement in control and cost exposure; the full comparative canary must still determine whether it improves scientific conversion.

External benchmark strategy

ASTRA has pinned adapters or protocols for SciCode, miniF2F, FrontierScience and AInsteinBench. SciCode evaluates code generation on scientist-curated research problems,⁵ FrontierScience tests expert-level scientific reasoning,⁶ and AInsteinBench targets coding agents inside scientific software repositories.⁷ AstaBench is a useful future umbrella because it spans literature, data, planning, tools, coding and search across more than 2,400 problems.⁸

⁵Tian et al., "SciCode: A Research Coding Benchmark Curated by Scientists," 2024.

⁶OpenAI, "Evaluating AI's ability to perform scientific research tasks," 2025.

⁷ByteDance Seed, "AInsteinBench," accessed July 2026.

⁸Allen Institute for AI, "AstaBench," accessed July 2026.

Benchmark discipline. Adapter readiness is not a score. Operational errors are excluded from scientific pass rates, architecture comparisons use frozen cases and budgets, and qualitative depth or novelty requires architecture-blinded expert review. The machine-readable release snapshot is tracked with the source repository.

Compact operating profile

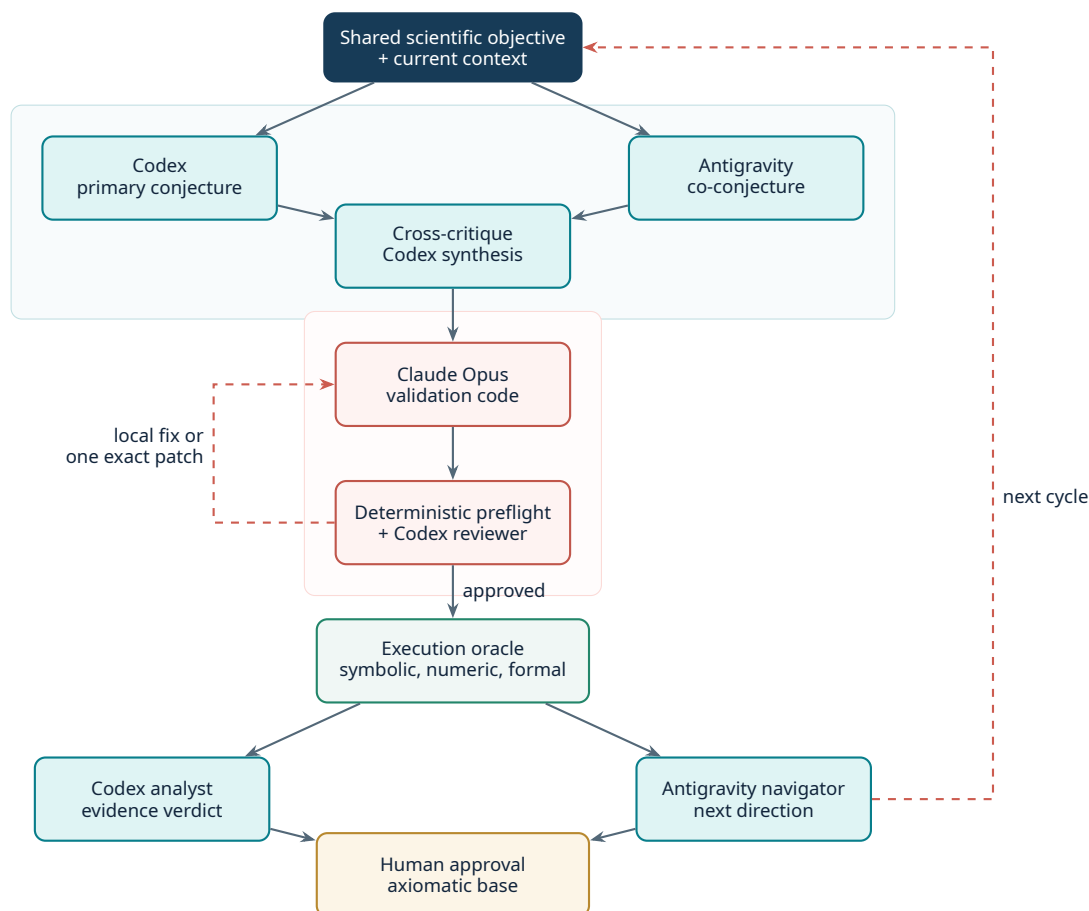
Resource dimension	ASTRA production profile
Foundation-model development	No model pretraining or fine-tuning is required; ASTRA orchestrates authenticated model services.
Model portfolio	Codex gpt-5.6-sol at xhigh, Antigravity/agy gemini-3.1-pro-high, and Claude Code claude-opus-4-8, with phase-specific roles and recorded bounded fallbacks.
Scientific compute	Local workstation plus one 16-core / 32-thread ASTRUM node with 29 GiB RAM and an RTX 3080; heavy validators route remotely.
Validation software	Primarily open scientific engines plus optional Mathematica and internal packages.
Repair exposure	Deterministic edits first; at most one bounded model patch in vNext.1.
Measured cost proxies	Model-call count, phase wall time, oracle time, repair attempts and evidence yield per call.
Unmeasured economics	No audited total-cost-per-accepted-result comparison with frontier systems yet.

Metrics for design-partner pilots

Product metric	Why it matters
Time and model calls to first reproducible test	Measures research-cycle compression and inference exposure
Independent reviewer rejection and recovery	Measures value of heterogeneous perspectives
False-acceptance rate on adversarial holdouts	Core assurance and trust metric
Cross-oracle reproduction rate	Tests portability and environmental independence
Human review time per accepted result	Tests whether traceability reduces expert burden
Accepted evidence per dollar or subscription-hour	Required before making a cost-superiority claim

Part II – System and algorithmic detail

Production architecture



Role separation and bounded correction are the central controls. Codex leads conjecture, synthesis, code review and evidence analysis. Antigravity contributes an independent conjecture and maintains research direction. Claude authors validation code and may supply one exact-edit repair. ASTRA first applies safe deterministic fixes and compile/import smoke checks. Deterministic engines, not model consensus, produce the evidence.

Interfaces and deployment

Layer	Current implementation
Interaction	Browser interface, command-line subprocess API and MCP tools for agents
Orchestration	Asynchronous Python pipeline with progress, timeouts, fallbacks and detached jobs
Models	Subscription-authenticated Codex, Claude Code and Antigravity CLIs; optional API providers
Execution	Local Python and optional remote Linux worker; engine-specific routing
Persistence	Reports, cycle state, research branches, benchmark results and human approval
Security posture	Credentials remain in local configuration; private execution is supported; enterprise controls remain a hardening priority

Fast and deep execution profiles

What can run in parallel. Independent validation cases, random seeds, parameter sweeps and cross-oracle replications can fan out across local workers and ASTRUM. Within a single scientific cycle, model roles remain ordered where one artifact depends on another. Subscription-backed model calls are kept sequential in comparative benchmarks so contention and quota timing do not become hidden confounders.

ASTRUM exposes 16 physical / 32 logical CPU threads, 29 GiB RAM and an RTX 3080. CPU-parallel SciPy, NumPy and symbolic workloads use the node when appropriate; GPU work routes only when the validator is genuinely GPU-shaped. The fast release profile uses deterministic regression and replay gates first. The deep profile runs frozen, long-horizon model trajectories and architecture-blinded expert scoring.

In the current four-worker smoke, eight deterministic local/ASTRUM runs all earned grade A, with 100% verdict accuracy and cross-oracle agreement. This demonstrates parallel release throughput and portability; it does not measure multi-model scientific depth.

The core algorithm

Inputs: shared objective G , established context K , current direction D , maximum code revisions R , execution policy O .

```

1.  $p_c \leftarrow \text{Codex.propose}(G, K, D)$ 
2.  $p_a \leftarrow \text{Antigravity.propose}(G, K, D)$  [parallel]
3.  $q_c \leftarrow \text{Codex.critique}(p_a)$ ;  $q_a \leftarrow \text{Antigravity.critique}(p_c)$ 
4.  $h \leftarrow \text{Codex.synthesize}(G, p_c, p_a, q_c, q_a)$ 
5.  $v_0 \leftarrow \text{Claude.translate}(G, h)$ 
6.  $(d, c) \leftarrow \text{Preflight.audit\_and\_compile}(v_0)$ 
7.  $v_1 \leftarrow \text{LocalRepair}(v_0, d)$  [safe patterns only]
8.  $r \leftarrow \text{Codex.review}(G, h, v_1, c)$ 
9. if  $r = \text{REVISE}$  and patchable:
     $p \leftarrow \text{Claude.exactPatch}(v_1, r)$ ;  $v_2 \leftarrow \text{ApplyBounded}(v_1, p)$ 
10.  $e \leftarrow \text{Oracle.execute}(v_2, O)$ 
11.  $s \leftarrow \text{Guard}(v_2, e)$ 
12.  $a \leftarrow \text{Codex.analyze}(G, h, v_2, e, s)$ 
13.  $n \leftarrow \text{Antigravity.navigate}(G, K, h, a)$ 
14.  $\text{HumanGate}(h, v_2, e, a, n)$ 

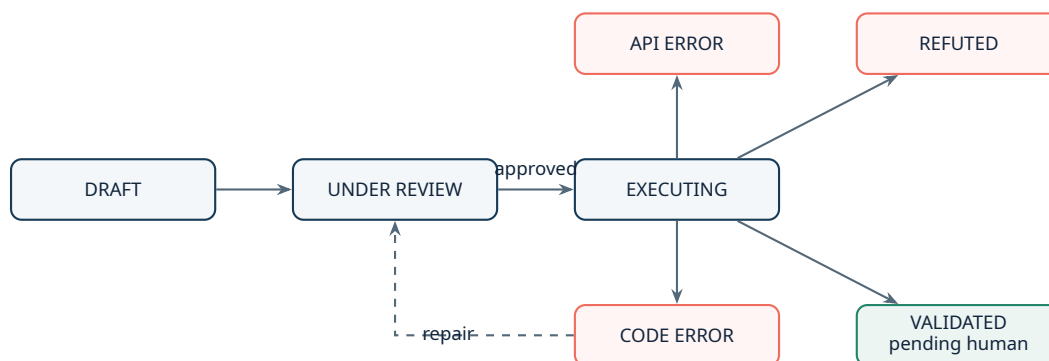
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Conservative decision semantics

The orchestration enforces several monotonic constraints:

- ▶ a crashed validator cannot become VALIDATED;
- ▶ an explicit FAIL cannot be promoted by explanatory prose;
- ▶ a printed PASS is evidence, not authority;
- ▶ missing optional engines fail explicitly;
- ▶ a reviewer can reject circular, unreachable or sampling-only proof logic;
- ▶ final admission to the axiomatic base requires a human decision.

State model



Interpretation boundary. In ASTRA, VALIDATED means supported by the recorded validator, execution and assumptions. It does not mean absolute truth. The evidence remains open to stronger proof, independent replication and expert review.

Scientific engine and extension model

Oracle portfolio

Engine class	Examples	Validation role
Symbolic	SymPy, SageMath, Maxima	Exact simplification, identities, derivatives, algebraic residuals
Logic / proof	Z3, Lean 4 + Mathlib	Counterexample search, satisfiability, kernel-checked theorems
Numerical	NumPy, SciPy, mpmath, QuTiP	ODEs, high-precision checks, quantum systems, sweeps
Domain packages	EinsteinPy, fluids, Pint	Curvature, fluid models, units and physical consistency
Independent CAS	Mathematica bridge, Cadabra	Cross-engine validation, tensor algebra, notebook workflows
Internal software	GR_python, pyWarpFactory, warp-bubble optimization	Reuse of proprietary or project-specific research capabilities

Extension contract

Any scientific package can become part of the oracle when a validator:

1. declares reproducible inputs, versions and assumptions;
2. imports the package in the selected environment;
3. computes a quantity that discriminates the claim from its negation;
4. exposes exact output, errors and tolerances;
5. includes reachable failure behavior;
6. preserves the script and result for review.

Agent control through MCP

ASTRA exposes operations for direct execution, complete cycles, progress probes, detached jobs, job polling and remote-worker status. A conversational agent can therefore plan a test and invoke ASTRA without manually copying code, while ASTRA retains the evidence and decision semantics.

Illustrative client instruction:

"Formulate two competing explanations for this curvature result. Use ASTRA to construct a validator with an exact symbolic leg and an independent numerical leg. Do not accept a printed PASS until the code, residuals and assumptions have been reviewed."

Enterprise productization priorities

- ▶ workspace authentication and role-based access;
- ▶ signed evidence bundles and immutable audit metadata;
- ▶ dependency and container pinning;
- ▶ job isolation and compute quotas;
- ▶ private model and on-premise connectors;
- ▶ team review queues and approval policy;
- ▶ observability, cost and latency dashboards;
- ▶ benchmark governance and release qualification.

Commercialization roadmap

Phase	Product objective	Evidence required before advancing
Now	Operational multi-model research loop, vNext.1 assurance and scientific oracle	Maintain 100% release-suite pass; harden maximum-model phase budgets; complete the frozen comparative canary; publish evidence snapshots
Pilot hardening	Deploy with a small number of design partners	Measured time-to-evidence, reviewer correction value, reproduction rate and expert satisfaction
Team product	Shared projects, governed approvals and scalable jobs	Repeat use, retained teams, security acceptance and clear economic buyer
Vertical modules	Domain-specific validators, packages and benchmark suites	Superior outcomes in selected high-value workflows
Platform expansion	Private deployment, partner engines and model marketplace	Reliable operations, integration demand and sustainable unit economics

Near-term capital and partnership priorities

1. **Product engineering:** packaging, secure execution, observability, collaboration and evidence management.
2. **Scientific calibration:** adversarial benchmarks, false-acceptance measurement and expert-reviewed domain suites.
3. **Design partnerships:** two or three workflows with high verification cost and measurable research-cycle value.
4. **Distribution:** model, compute, scientific-software and research institution integrations.

Risks and mitigations

Risk	Why it matters	Mitigation direction
Validator mismatch	Code may test a weaker or different claim	Independent code review, human gate, benchmarked failure patterns
Model latency and quota	Maximum-quality cycles can take tens of minutes	Deterministic repair, one bounded patch, phase budgets, caching, detached runs and ASTRUM fan-out
False confidence	Users may overread VALIDATED	Explicit semantics, evidence display, independent engines, approval policy
Scientific breadth	No system is expert in every domain	Vertical focus, package adapters, domain reviewers and scoped claims
Enterprise security	Generated code executes with real resources	Isolation, allowlists, containers, credentials policy and audit logs
Go-to-market diffusion	"Science" is too broad as an initial segment	Start with measurable design-partner workflows and a specific buyer

The next proof points

- ▶ rerun the maximum-model one-cell diagnostic after phase-budget hardening, then complete the frozen vNext.1 comparison against baseline and vNext.0;
- ▶ expand adversarial holdouts beyond the four deterministic repair patterns;
- ▶ publish scored SciCode, FrontierScience, miniF2F and AInsteinBench subsets with operational failures reported separately;
- ▶ quantify expert time, model calls and compute cost per accepted or refuted claim;
- ▶ identify one vertical where ASTRA becomes a paid part of the weekly R&D workflow.

Engagement proposal

The product thesis in one sentence

ASTRA is the verification and coordination layer that allows organizations to use frontier AI in scientific work while preserving falsifiability, reproducibility and human control.

For a prospective client

Select one consequential workflow with:

- ▶ a clear scientific decision and current baseline process;
- ▶ existing code, equations, simulations or internal packages;
- ▶ measurable verification effort or rework;
- ▶ an expert who can judge whether ASTRA preserves the real claim.

A pilot should compare baseline and ASTRA-assisted work on time-to-evidence, reproducibility, review effort and false acceptance.

For an investor or VC

The immediate diligence questions are concrete:

1. Does heterogeneous review materially improve generated validators under equal budgets?
2. Which customer segment has the highest cost of unverifiable AI output?
3. Can the evidence architecture become a repeatable team workflow?

4. Which data, adapters and calibration assets compound with usage?
5. Can private deployment and scientific specialization support durable economics despite frontier-model dependence?

For a strategic partner

ASTRA can provide a governed research workflow around:

- ▶ frontier models that need credible scientific demonstrations;
- ▶ compute platforms that want traceable high-value workloads;
- ▶ scientific-software vendors seeking agent-native interfaces;
- ▶ research organizations building reproducible AI-assisted methods.

Proposed next step. Define a 4–8 week design-partner study around one scientific workflow, one baseline, one reproducibility target and an agreed set of acceptance metrics. The goal is not a demo; it is evidence that ASTRA changes a real R&D decision.

Astrum Drive value proposition. ASTRA turns frontier-model capability into a governed scientific workflow that Astrum Drive can own, calibrate and integrate. The near-term asset is not a claim to have built a larger model; it is the control plane, validator library, benchmark discipline, package graph and accumulated evidence needed to make powerful models useful in consequential physics and engineering work.

ASTRA

From plausible answers to reviewable evidence.

Nelson Bolívar | Founder and Lead Architect | Astrum Drive
Investor, client and strategic partner discussions | July 2026